CV

CostVision

Manufacturing Should-Cost Intelligence Platform

SHOULD-COST ANALYSIS REPORT

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Bottom-Up Manufacturing Cost Model · Fully Traceable Rate Data

Bosch IPB2.0 ECU PCBA

Commodity: PCBA · Currency: GBP · FX Rate: 1.0000 to GBP · Operations: 4 · Region: CN

Total Should-Cost	Material	Process	Labour	Margin
£185.32	**72.1%**	**11.0%**	**1.1%**	**7.4%**

Model Confidence: **Low** · 4 traced operations · 9 data points auditable

Key Assumptions

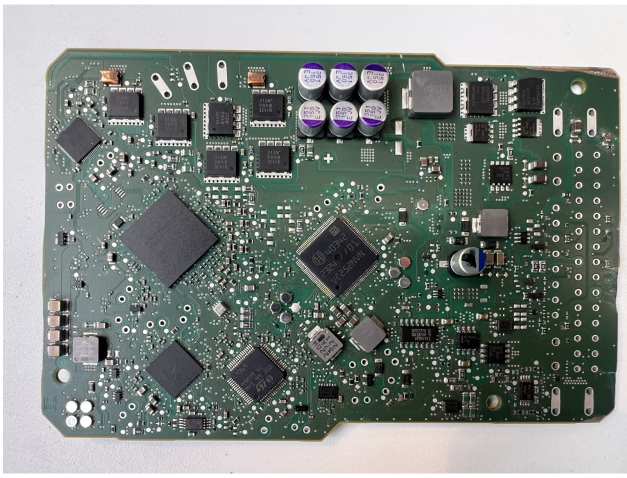
Annual volume: 150,000 · Region: CN · Commodity: pcba
Alloy / material: Virtual / Pass-through · Net weight: 0.000 kg
Material utilisation: 100% · Overhead: 9% of factory base · Margin: 8% of subtotal · Operations: 4

Source Photographs

1 image the costing was built from

Every component identification in the bill of materials traces to a package marking legible in one of these images. Lines flagged as estimated could not be read from any of them.

Top side



§1 — 8-Bucket Cost Breakdown

Cost Bucket	Amount (GBP)	% of Total	Cost Mix Bar
1. Raw Material	£133.56	72.1%	
2. Process (Machine)	£20.38	11.0%	
3. Direct Labour	£1.99	1.1%	
4. Tooling (amortised)	£0.31	0.2%	
5. Packaging	£0.73	0.4%	
6. Logistics	£0.57	0.3%	
Factory Cost	**£157.53**	**85.0%**	
7. Overhead (SG&A)	£14.06	7.6%	
Subtotal	**£171.59**	**92.6%**	
8. Supplier Margin	£13.73	7.4%	
TOTAL SHOULD-COST	**£185.32**	**100.0%**	

§2 — Commercial Parameters

Overhead Rate	**9.0%**	Supplier Margin	**8.0%**
Packaging / part	**£0.73**	Logistics / part	**£0.57**
Total Tooling Cost	**£46000.00**	Amortisation Volume	**150,000 parts**

§3 — Material Detail

Parameter	Value	Unit	Notes
Material ID	**mat-virtual**	ID	
Grade / Specification	**Virtual / Pass-through**		Placeholder for directCost modules (painting, BIW, PCB, PCBA). Price is irrelevant — directCost overrides. | Regional adj. ×0.880 (other)
Region	**China**		
Net Finished Weight	**0.0000 kg**	kg	Weight in finished part
Direct Material Cost	**£133.56**	GBP	Bypasses weight-based model
Data Confidence	**Medium**		
Effective Date	**2026-06-14**		

3B Material Line Items (20 lines)

Ref	Description	Qty	Unit GBP	Ext GBP	Source / basis
U-MCU	Renesas R7F702300B RH850/U2A automotive MCU	1	£15.00	**£15.00**	
U-ASIC1	Bosch 40342/01 motor-control ASIC (captive)	1	£10.00	**£10.00**	
U-ASIC2	Bosch 40341/01 ASIC (captive)	1	£7.00	**£7.00**	
U-ASIC3	Bosch 2302701 ASIC (captive)	1	£4.00	**£4.00**	
U-SBC	ST L9369 multi-channel driver / SBC	1	£5.00	**£5.00**	
U-BGA2	Secondary safety MCU / ASIC (BGA)	1	£8.00	**£8.00**	
U-MEM	External memory (BGA)	1	£3.00	**£3.00**	
Q1-6	Q142E power stage / dual half-bridge	6	£3.00	**£18.00**	
U-CAN	NXP TJA1463A CAN-FD SIC transceiver	2	£1.20	**£2.40**	
U-71H	Bosch 71H740 driver / power-path (captive)	3	£1.50	**£4.50**	
U-76E2	Bosch 76E240 device (captive)	1	£1.80	**£1.80**	
U-76E8	Bosch 76E840 device (captive)	1	£1.80	**£1.80**	
U-7S1R	Bosch 7S1R540H power device (captive)	2	£1.60	**£3.20**	
C-BULK	Polymer alu cap 150 uF 35 V (351 150 EJV)	6	£0.40	**£2.40**	
L-PWR	Shielded power inductor 0.47-10 uH	8	£0.60	**£4.80**	

Ref	Description	Qty	Unit GBP	Ext GBP	Source / basis
R/C-SMD	MLCC + thick-film resistors, AEC-Q200	620	£0.01	**£5.58**	
D/Q-SML	Small-signal semis (SOT/SOD/QFN)	90	£0.09	**£8.10**	
Y1	Crystal / oscillator	1	£0.45	**£0.45**	
J1-2	Automotive press-fit header (main + motor)	2	£2.50	**£5.00**	
TIM	Thermal interface pad (power stages)	8	£0.12	**£0.96**	
	LINE ITEM TOTAL			**£110.99**	

Lines total £110.99 against a £133.56 material bucket — the £22.57 balance is carried outside the itemisation (bare board, yield/rework allowance, coating).

§4 — Operations Detail

4A Machine Operations

Operation	Machine Class	Rate / hr	Cycle (min)	Parts / Cycle	OEE %	Machine Cost
SMT Placement & Reflow	High-Speed SMT Line (Fuji/Juki/ASM)	£78.38	5.51	1	82.0%	**£8.78**
TH Insertion & Hand Soldering	Assembly Workbench	£0.36	0.40	1	100.0 %	**£0.00**
BGA X-Ray Inspection (100% offline)	X-Ray BGA Inspection Cell (2D/3D AXI)	£47.63	7.92	1	90.0%	**£6.99**
ICT / Functional Test	ICT Bed-of-Nails Test System (Automotive)	£58.39	4.50	1	95.0%	**£4.61**
TOTAL						**£20.38**

4B Labour Detail

Operation	Labour Grade	Rate / hr	Man ning	Lab. min	Eff %	Labour Cost	Op Total
SMT Placement & Reflow	SMT / Electronics Operator	£6.50	1	5.51	100.0%	£0.60	**£9.38**
TH Insertion & Hand Soldering	SMT / Electronics Operator	£6.50	1	0.40	100.0%	£0.04	**£0.05**
BGA X-Ray Inspection (100% offline)	SMT / Electronics Operator	£6.50	1	7.92	100.0%	£0.86	**£7.84**
ICT / Functional Test	SMT / Electronics Operator	£6.50	1	4.50	100.0%	£0.49	**£5.10**
TOTAL						**£1.99**	**£22.36**

§5 — Machine Rate Buildup

Assembly Workbench [bench-assembly] · Confidence: Low · Assembly bench — low machine rate; cost primarily driven by labour | Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh

Cost Component	Annual Cost (GBP)	Rate/hr @85% util	Notes
Depreciation	£275.00	£0.08	4,000 hr/yr available
Maintenance	£110.00	£0.03	
Energy	£91.30	£0.03	
Floor Space	£440.00	£0.13	
Indirect Support	£275.00	£0.08	
Finance Cost	£27.50	£0.01	
TOTAL — Assembly Workbench	£1218.80	£0.36	Effective hours: 3400/yr

High-Speed SMT Line (Fuji/Juki/ASM) [smt-high-speed-line] · Confidence: Low · High-speed SMT line 80000+ CPH. Automotive EMS benchmark. Target £150/hr. Jun 2026 | Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh

Cost Component	Annual Cost (GBP)	Rate/hr @82% util	Notes
Depreciation	£99000.00	£30.18	4,000 hr/yr available
Maintenance	£44000.00	£13.41	
Energy	£16739.13	£5.10	
Floor Space	£16500.00	£5.03	
Indirect Support	£55000.00	£16.77	
Finance Cost	£25850.00	£7.88	
TOTAL — High-Speed SMT Line (Fuji/Juki/ASM)	£257089.13	£78.38	Effective hours: 3280/yr

X-Ray BGA Inspection Cell (2D/3D AXI) [xray-bga-inspection] · Confidence: Low · 2D/3D automated X-ray inspection for BGA solder joints. EMS automotive benchmark. Target £90/hr. Jun 2026 | Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh

Cost Component	Annual Cost (GBP)	Rate/hr @82% util	Notes
Depreciation	£66000.00	£20.12	4,000 hr/yr available
Maintenance	£24750.00	£7.55	
Energy	£7608.70	£2.32	
Floor Space	£11000.00	£3.35	
Indirect Support	£35750.00	£10.90	
Finance Cost	£11110.00	£3.39	
TOTAL — X-Ray BGA Inspection	£156218.70	£47.63	Effective hours: 3280/yr

Cell (2D/3D AXI)

ICT Bed-of-Nails Test System (Automotive) [ict-automotive] · Confidence: Low · In-circuit test fixture for IATF 16949 automotive boards. Target £110/hr. Jun 2026 | Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh

Cost Component	Annual Cost (GBP)	Rate/hr @82% util	Notes
Depreciation	£71500.00	£21.80	4,000 hr/yr available
Maintenance	£30250.00	£9.22	
Energy	£9130.43	£2.78	
Floor Space	£13750.00	£4.19	
Indirect Support	£46750.00	£14.25	
Finance Cost	£20130.00	£6.14	
TOTAL — ICT Bed-of-Nails Test System (Automotive)	£191510.43	£58.39	Effective hours: 3280/yr

How to read the machine rate

Rate/hr = (annual depreciation + maintenance + energy + floor + indirect + finance) ÷ effective hours, where effective hours = available hours × utilisation.
Effective hours encode the shift pattern: a lower hours-base raises £/hr and a higher one lowers it. A challenge from a supplier will target this divisor and whether depreciation is machine-only or full-cell (machine + dosing furnace + trim + robots/extraction) — state your basis when defending the number.

§6 — Rate Traceability

Field	Value	Unit	Source / Reference	Rate ID	Conf.
rawMaterial.directCost	133.5586	£	Pre-computed by commodity module	mat-virtual	Medium
SMT Placement & Reflow.machineRatePerHr	78.3808	£/hr	High-speed SMT line 80000+ CPH. Automotive EMS benchmark. Target £150/hr. Jun 2026 | Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh	smt-high-speed-line	Low
SMT Placement & Reflow.labourRatePerHr	6.5000	£/hr	Regional benchmark China — 2026 Q2	lab-cn-electronics	Low
TH Insertion & Hand Soldering.machineRatePerHr	0.3585	£/hr	Assembly bench — low machine rate; cost primarily driven by labour | Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh	bench-assembly	Low
TH Insertion & Hand Soldering.labourRatePerHr	6.5000	£/hr	Regional benchmark China — 2026 Q2	lab-cn-electronics	Low
BGA X-Ray Inspection (100% offline).machineRatePerHr	47.6277	£/hr	2D/3D automated X-ray inspection for BGA solder joints. EMS automotive benchmark. Target £90/hr. Jun 2026 | Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh	xray-bga-inspection	Low
BGA X-Ray Inspection (100% offline).labourRatePerHr	6.5000	£/hr	Regional benchmark China — 2026 Q2	lab-cn-electronics	Low

Field	Value	Unit	Source / Reference	Rate ID	Conf.
ICT / Functional Test.machineRatePerHr	58.3873	£/hr	In-circuit test fixture for IATF 16949 automotive boards. Target £110/hr. Jun 2026 | Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh	ict-automotive	Low
ICT / Functional Test.labourRatePerHr	6.5000	£/hr	Regional benchmark China — 2026 Q2	lab-cn-electronics	Low

§7 — Should-Cost with Uncertainty

Monte-Carlo · Low confidence

Estimate (± band)	P10 · optimistic	P50 · median	P90 · conservative	Band	CV
£185.32 ± 23.8%	£144.10	£181.43	£232.19	wide	19%

Range reflects how well each cost driver is known (High / Medium / Low confidence). Tighten it by attaching CAD, a BOM, or logging an actual quote.

Why confidence is Low

Main band drivers: 8 rate(s) at Low confidence; tooling amortisation carries the widest per-bucket spread.

What's excluded from this unit cost

One-time NRE — tooling / die, fixtures and CNC programming — is amortised separately, not in this unit cost.
Import duty and international freight (regional table is Ex-Works).
Formal embodied-carbon reporting (figures are indicative cradle-to-gate — replace with supplier EPDs).

§8 — Sensitivity Analysis

± 10% driver swing · ranked by £ impact

Cost Driver	Base	-10% cost	+10% cost	Range £
Direct material cost	£133.56	-8.5%	+8.5%	**£31.45**
Overhead %	9%	-0.8%	+0.8%	**£3.04**
Supplier Margin %	8%	-0.7%	+0.7%	**£2.75**
SMT Placement & Reflow: Cycle Time	0.09hr	-0.6%	+0.6%	**£2.21**
SMT Placement & Reflow: Machine Rate	£78.38/hr	-0.6%	+0.6%	**£2.07**
BGA X-Ray Inspection (100% offline): Cycle Time	0.13hr	-0.5%	+0.5%	**£1.85**
BGA X-Ray Inspection (100% offline): Machine Rate	£47.63/hr	-0.4%	+0.4%	**£1.64**
ICT / Functional Test: Cycle Time	0.08hr	-0.3%	+0.3%	**£1.20**

§9 — Regional Cost Comparison

Ex-Works · per-region should-cost · vs China

Region	Material	Process	Labour	Tooling	Overhead	Ex-Works	Logistics	Total	vs China
United Kingdom (GBP)	£151.77	£37.05	£7.15	£0.56	£18.75	**£215.28**	£0.39	**£230.44**	+24%
Germany (EUR)	£156.32	£38.90	£11.55	£0.59	£21.76	**£229.13**	£0.45	**£245.29**	+32%
France (EUR)	£154.81	£34.09	£8.30	£0.51	£19.80	**£217.51**	£0.45	**£232.94**	+26%
Spain (EUR)	£151.77	£32.60	£5.24	£0.49	£17.23	**£207.33**	£0.45	**£222.07**	+20%
Poland (PLN)	£147.22	£26.68	£3.25	£0.40	£14.40	**£191.94**	£0.47	**£205.60**	+11%
Turkey (TRY)	£136.59	£22.23	£1.81	£0.33	£11.98	**£172.94**	£0.51	**£185.33**	+0%
China (CNY)	£133.56	£20.38	£1.99	£0.31	£14.06	**£170.29**	£0.57	**£185.32**	Base
India (INR)	£136.59	£19.27	£1.26	£0.29	£10.81	**£168.23**	£0.59	**£180.23**	-3%
Mexico (MXN)	£144.18	£22.23	£2.09	£0.33	£12.56	**£181.40**	£0.53	**£194.29**	+5%
United States (USD)	£151.77	£31.49	£9.39	£0.47	£17.50	**£210.63**	£0.49	**£225.55**	+22%

Per-region should-cost using regional labour, energy and rate benchmarks. Tooling held fixed. Ex-Works — excludes import duty + international freight. Indicative — confirm with RFQ.

§10 — Embodied Carbon

0.00 kgCO2e · 0.00 /kg · cradle-to-gate

Material	**0.00 kgCO2e**	Material factor	**3.00 kg/kg (Generic (assumed))**
Process	**0.00 kgCO2e**	Grid intensity	**0.550 kg/kWh · 0.00 kWh**
Logistics	**0.00 kgCO2e**	Total	**0.00 kgCO2e**

Indicative cradle-to-gate factors — replace with supplier EPDs for formal reporting.

§11 — Cost Intelligence Insights

2 findings · ~1% combined saving

[CRITICAL] Material is the dominant cost driver

Finding	Raw material at 72.1% of total exceeds the pcba benchmark ceiling of 70%. Material is controlling this part's economics.
Impact	High · up to 1% potential saving
Benchmark	Material %: yours 72.1% vs industry 45–70%
Action 1	Re-cost the BOM at the target volume — line items quoted at prototype price carry the largest error
Action 2	Consolidate part numbers across the passive families (a single 0402 100nF X7R across the board)

[WARNING] Low OEE on "SMT Placement & Reflow"

Finding	OEE of 82% is below the 90% benchmark. Every 5% OEE improvement on this operation saves ~£1.019/part.
Impact	Medium · up to 1% potential saving
Action 1	Implement SMED (Single-Minute Exchange of Die) to reduce changeover losses
Action 2	Schedule preventive maintenance to reduce unplanned downtime

§12 — Cost-Reduction Opportunities (ranked by saving)

4 opportunities · combined £13.16/part (~7%)

Ranked on money per part against this costing. The combined figure is the root-sum-square of the top three, capped at 40% — overlapping actions do not save twice, so the column is deliberately not summed.

#	Category	Action	Why it is on the list	Save/part	%	Timeframe	Risk
1	Design & Geometry	**Design Review for DFM Compliance**	Formal DFM review with manufacturing engineering to identify part features that increase cost without functional benefit.	**£9.27**	5.0%	Quick Win	Low
2	Quality & Inspection	**Right-Size Inspection and Test**	Inspection/test operations carry 7.0% of part cost as separate steps. Capable processes earn reduced inspection — that is what capability data is for.	**£7.41**	4.0%	Medium Term	Low
3	Commercial & Sourcing	**Raw-Material Price Indexation Clause**	With material at 72.1% of cost, the quote embeds the supplier's hedge against metal/resin volatility. An index-linked clause removes that risk premium from the piece price.	**£5.56**	3.0%	Quick Win	Low
4	Commercial & Sourcing	**Annual Volume Re-Commitment for Better Pricing**	Provide 12-month rolling volume forecast to supplier to secure better unit pricing and tooling amortisation.	**£5.56**	3.0%	Quick Win	Low

§13 — Opportunity by Category

3 categories with an actionable lever

Category	Opportunities	Best single action	Best save/part	Category total (indicative)
Design & Geometry	1	Design Review for DFM Compliance	**£9.27**	£9.27
Quality & Inspection	1	Right-Size Inspection and Test	**£7.41**	£7.41
Commercial & Sourcing	2	Raw-Material Price Indexation Clause	**£5.56**	£11.12

§15 — Implementation Roadmap

Quick Wins — Implement Immediately

- Raw-Material Price Indexation Clause
- Design Review for DFM Compliance
- Annual Volume Re-Commitment for Better Pricing